

Scientific Data Composition as a Capability-Shaping Mechanism for Foundation Models: Evidence from Biological Continued Pretraining

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Highlights

- A training-free re-analysis of one 26B MoE lineage isolates the effect of biological continued pretraining (CPT) from tokenizer changes and from fine-tuning.
- Biological CPT does not cause catastrophic forgetting; measured general-knowledge performance under likelihood evaluation *rises* (MMLU +13 pts) alongside code generation (MBPP pass@1 $0.33 \rightarrow 0.63$).
- CPT also shifts the observable reasoning trajectory on chain-of-thought tasks 41% shorter and near-revision-free while preserving accuracy — an effect invisible to accuracy metrics.
- A consistent CPT-lifts / SFT-narrows division of labor recurs across four axes, reframing biological sequence as structured scientific data that reshapes a model’s capability profile.
- A controlled seven-model experiment (varying only the CPT data mixture) confirms this causally: raising biological share to 50% preserves general ability (MMLU -0.6 pt) while biology rises monotonically — data mixture is a general-preserving lever on capability.
- A small-model, matched-compute causal probe (GPT-2, 3 seeds) finds that biological CPT *reproducibly lifts* paraphrase detection (PAWS-en, +3.1 pt) — the direct structural counterpart of protein-homology transfer — alongside a real cost to long-range discourse modeling, an honestly reported partial, not uniform, transfer.
- The general-preserving pattern holds across a $10\times$ scale range (2.3B–26B); an explicit replay schedule has nothing to recover (uniform mixing was already general-preserving); and router analysis shows full CPT reorganizes expert routing while LoRA-scale mixture ratio lifts capability without a matching routing dose-response — two dissociable mechanisms.

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The Bigger Picture.

As foundation models mature, the lever practitioners reach for less often is architecture and more often *what the model is trained on* — yet the composition of a pretraining corpus is usually treated as a budget to be minimized (how much of a specialized domain can we afford without hurting everything else) rather than a dial that can be turned deliberately to engineer a target capability profile. We test whether training-data composition is such a dial, using biological sequence — DNA and protein, which share almost no surface statistics with natural language — as a first, deliberately extreme case study. The usual assumption is that adapting a general-purpose model to so foreign a domain must come at a cost: teach it biology and it forgets how to reason about everything else. Comparing snapshots of the same model before and after biological training, without any new training of our own, overturns that intuition: feeding the model raw biological sequence left its general knowledge and code-writing ability intact and even changed how it reasons, producing shorter, less self-revising chains of thought. A seven-model controlled follow-up, in which we retrain the same base while varying only the data mixture, confirms this is a genuine dial rather than a property of one checkpoint: more biological data buys more scientific capability at almost no cost to general ability, and we show the same dial holds across a ten-fold range of model scale. A smaller, sharper experiment asks the question in reverse — does biological training feed anything back into ordinary language understanding — and finds that it does, reproducibly, on the one task that theory predicts it should, though not without a cost elsewhere that we report honestly rather than smooth over. Together the results reframe biological sequence not as a domain to be tolerated but as the first worked example of a broader principle: structured scientific data — biology today, and by extension code, mathematics, and chemistry — is a lever for engineering what a foundation model can do.

Abstract

Foundation-model capability is usually engineered through architecture and scale; we ask whether training-data *composition* is itself a controllable dial for shaping what a model can do, using biological sequence — DNA and protein, which share almost no surface statistics with natural language — as a first, deliberately extreme test case. The standard intuition is that continued pretraining (CPT) on so narrow, out-of-distribution a corpus must trade away a general-purpose model’s broad competence — the “alignment tax” or catastrophic-forgetting view. We test this directly, without any new training, by re-analyzing three checkpoints from a single lineage of a 26B-parameter Mixture-of-Experts model (Gemma-4-26B-A4B): the instruction-tuned base, the same model after biological CPT (8.7B tokens of DNA, protein, and biomedical text), and after subsequent supervised fine-tuning (SFT). Across three independent capability axes — general knowledge/reasoning (MMLU, ARC, HellaSwag), code generation (MBPP), and biomedical knowledge (BixBench) — we find that biological CPT does *not* degrade the model; it **lifts** it: MMLU rises +13 points under likelihood evaluation, MBPP pass@1 nearly doubles (0.33→0.63), and BixBench discrimination rises sharply (MCC 0.23→0.92). The single measured regression is truthfulness (TruthfulQA −8.8 points), a small and interpretable domain drift. A clean vocabulary-expansion ablation (<0.4 pt on every general metric) confirms the gains are attributable to CPT, not tokenizer changes. Crucially, subsequent SFT *narrows* the model back: all three axes fall to near-base levels, revealing a consistent division of labor — **CPT re-organizes and lifts the shared capability substrate; SFT cashes it out onto target tasks.** We argue this reframes biological sequence not as a competitor for a foundation model’s capacity but as a form of structured scientific data that reshapes its capability profile, and that CPT and SFT should be budgeted as complementary rather than substitutable stages. We then confirm the mechanism causally: continuing pretraining of the same base seven times under one fixed recipe, varying only the data mixture (0/5/20/50% biological share plus a

protein/DNA composition ablation), shows that raising the biological share to 50% leaves the general axis essentially unchanged (MMLU -0.6 pt, GSM8K -1.2 pt) while biological capability rises monotonically (remote-homology MCC flips negative to positive; BixBench saturates), and the *type* of scientific data selects which capability is amplified. Mixture ratio is thus a controllable, general-preserving lever on the capability profile. Finally, we ask the reverse question at a scale with capability headroom the 26B model lacks: does biological CPT feed anything back into ordinary language understanding? A matched-compute continuation experiment on GPT-2 (124M), holding token budget, steps, and optimizer identical between a biology-continued and a text-continued arm across three seeds, finds a real, reproducible transfer to the direct structural counterpart of protein-homology detection — paraphrase detection (PAWS-en, $+3.1$ points, consistent in sign across all three seeds) — alongside a genuine cost to long-range discourse modeling (LAMBADA, -14.6 points), a partial rather than uniform dissociation that we report honestly. We close three remaining gaps directly. The general-preserving pattern of the mixture experiment holds across a $10\times$ range of model scale (2.3B–26B, MMLU delta within noise at every size). An explicit end-of-training text replay schedule, tested against the uniformly-mixed baseline at matched tokens, changes nothing on the general axis (already flat) and mildly reduces biology-axis performance — uniform mixing was not forgetting anything a replay buffer could recover. And router-level analysis across every checkpoint in this paper shows that full-parameter CPT visibly reorganizes expert-routing allocation between biological and linguistic inputs, while the LoRA-scale mixture sweep shows no such routing dose-response despite a clear capability dose-response — capability shaping and routing reorganization are dissociable mechanisms that converge only at large training budgets. All checkpoints, evaluation code, and per-example outputs are public.

Keywords: foundation models, continued pretraining, catastrophic forgetting, data-centric AI, mixture-of-experts, capability transfer, structured scientific data.

1 Introduction

The dominant intuition about continued pretraining (CPT) on a narrow domain corpus is subtractive: pouring out-of-distribution tokens — raw protein and DNA sequence, which share almost no surface statistics with natural language — into a general-purpose model is expected to erode its broad competence, a phenomenon variously described as the *alignment tax* or *catastrophic forgetting* [1, 2]. Under this view, domain adaptation is a trade: one buys biological ability at the cost of general ability, and the practitioner’s job is to minimize the damage.

This paper argues that, for a modern Mixture-of-Experts foundation model, the intuition is wrong — or at least incomplete — and uses biological sequence as a first, deliberately extreme test case for a broader claim: that training-data *composition* is a controllable lever for shaping a foundation model’s capability profile, not merely a cost to be minimized. We show, by *re-analysis of existing checkpoints and without any new training*, that biological CPT on a 26B-parameter model does not merely avoid catastrophic forgetting; measured performance *rises* on capability axes that have nothing to do with biology, including general knowledge under likelihood evaluation (MMLU [3]) and code generation (MBPP). We then show that the subsequent supervised fine-tuning (SFT) stage, far from adding capability, *narrows* the model back toward its base competence on held-out axes. The picture that emerges is a clean division of labor between the two training stages, and it suggests a reframing of what biological data *is* to a foundation model: not a domain to be tolerated, but the first worked example of a design surface — data composition — that Parts II–IV of this paper show is controllable, causal, and mechanistically dissociable from mere routing change.

Our substrate is a Mixture-of-Experts model [4–6] (Gemma-4-26B-A4B [7]); the analysis is deliberately training-free, re-using an existing CPT/SFT lineage [2] so that every capability difference

is attributable to a single stage. This complements router-level interpretability of the same model family [8–10] by asking not *where* computation is organized but *how* the training data reshapes what the model can do.

Related work. Our claim sits at the intersection of three literatures. *Continued pretraining and forgetting*: domain-adaptive CPT is standard practice [2, 11, 12], but is widely reported to risk catastrophic forgetting of general ability [1, 13], motivating replay and balancing schemes. Most of this evidence is on dense models and measures forgetting as a cost to be minimized; we instead find a net *gain* on a sparse MoE model and attribute it to the structural regularity of the pretraining corpus rather than its factual content. *Biological foundation models*: protein and genomic models — ESM-2 [14], ProtT5 [15], DNABERT-2 [16], Nucleotide Transformer [17], the Evo family [18, 19], and cross-modal LucaOne [20], as well as LLMs adapted to biology [21–23] — are typically evaluated only on in-domain tasks, leaving open whether biological training helps or harms general competence. *Data-centric AI and scaling*: work on pretraining-data composition and its effect on capability [24, 25] argues that *what* a model is trained on shapes *what* it can do; we provide a controlled, single-lineage instance of this principle where the manipulated variable is a scientific-sequence corpus and the readout spans knowledge, code, and reasoning behavior. To our knowledge no prior work reports that biological CPT raises measured general and coding benchmark performance, nor that it measurably shifts the observable reasoning trajectory.

Contributions. (i) A training-free re-analysis protocol over a single checkpoint lineage that isolates the effect of biological CPT from tokenizer changes and from SFT. (ii) Evidence that biological CPT lifts, rather than degrades, general and code capability, with a single interpretable regression (truthfulness). (iii) A consistent CPT-lifts / SFT-narrows division of labor observed across four independent evaluation axes. (iv) A data-centric reframing: biological sequence as structured scientific data that reshapes a foundation model’s capability profile. (v) A controlled seven-model mixture experiment that turns the observational claim causal: varying only the CPT data mixture, general ability is preserved across a 0–50% biological share while biological ability rises monotonically, and data *composition* selects which capability is amplified. (vi) A small-model, matched-compute causal probe of the reverse direction, showing that biological CPT reproducibly improves natural-language paraphrase detection — the direct counterpart of the protein-homology transfer that motivates this line of work — while honestly reporting a real cost to long-range discourse and a scale at which character-level structural probes are simply unmeasurable. (vii) Three robustness and mechanism checks that close the remaining gaps: the general-preserving pattern holds across a $10\times$ model-scale range; an explicit replay schedule has nothing to recover because uniform mixing was never forgetting anything at this budget; and router-level analysis dissociates capability shaping (present at LoRA scale) from routing reorganization (present only at full-CPT scale), a genuine mechanistic nuance rather than a clean confirmation of either extreme.

2 Setup: one lineage, three checkpoints

Our analysis rests on a single training lineage, so that every capability difference we report can be attributed to a specific training stage rather than to confounds of architecture, tokenizer, or data version.

Base model. All checkpoints derive from Gemma-4-26B-A4B, a Mixture-of-Experts transformer (128 experts per layer, top-8 routing, ≈ 3.8 B active parameters). We use the instruction-tuned release

as the origin of the lineage; the biological CPT and SFT checkpoints are those of OmniGene-4 [8].

The three (four) checkpoints. The lineage is

$$\text{it} \xrightarrow{\text{vocab-expand}} \text{it-bio} \xrightarrow{\text{bio-CPT}} \text{BioCPT} \xrightarrow{\text{SFT}} \text{BioCPT+SFT}.$$

(1) Base (it): the original instruction-tuned model, vocabulary 262,144. **(2) Base (it-bio):** identical weights with the vocabulary expanded to 290,172 by adding 28,028 biological BPE tokens (DNA/protein), but *no further training*; this serves as a clean control that isolates the cost of tokenizer expansion alone. **(3) BioCPT:** it-bio after continued pretraining on 8.7B tokens of mixed biological corpus (DNA, protein sequence, and biomedical literature text), with LoRA ($r=64$) over attention, MLP, and the MoE router, embeddings unfrozen; merged to full weights. **(4) BioCPT+SFT:** BioCPT after supervised fine-tuning on biological instruction data (a dual-head structural variant, “v5”); merged to full weights. We verified the lineage by direct weight comparison: it and it-bio share router/expert weights bit-for-bit (mean abs. diff. 0), differing only in embeddings, while BioCPT’s router and expert weights have measurably moved.

Evaluation. We use the `lm-evaluation-harness` (v0.4.12) for all general and coding benchmarks, at community-standard few-shot settings (MMLU 5-shot, ARC-Challenge 25-shot, HellaSwag 10-shot, TruthfulQA-mc2 0-shot; HumanEval 0-shot, MBPP 3-shot). Multiple-choice tasks are scored by loglikelihood, which is robust to a model’s chat-vs-completion output format — a property we exploit deliberately, since our lineage mixes an instruction-tuned base with a completion-style CPT checkpoint. Biomedical discrimination (BixBench) is likewise scored by loglikelihood of the answer choices, and reasoning-behavior diagnostics (§3) use a shared 8-shot chain-of-thought scaffold. All runs are on a single 96 GB GPU in bfloat16; the full model (≈ 52 GB) fits on one card, and the entire study required no training.

Honest scope. Two evaluation regimes are format-sensitive and we flag them up front. HumanEval 0-shot (bare function-completion) is unfair to instruction/SFT checkpoints and we exclude it from conclusions, reporting MBPP 3-shot as the fair coding metric. Zero-shot protein *homology* is not measurable by any simple probe we tried (forced Yes/No loglikelihood is near-random for all checkpoints; free generation is format-confounded); we therefore treat sequence-homology ability separately and do not force it into the transfer analysis.

3 Part I — Observational re-analysis of one lineage

The central finding is a single pattern that recurs across every capability axis we measured: biological CPT *lifts* the model, and subsequent SFT *narrows* it back (Figure 1, Figure 3). We present the three completed axes, then the reasoning-behavior diagnostics.

3.1 Vocabulary expansion is free

Before attributing anything to CPT we rule out the tokenizer. The original model (it) and the vocabulary-expanded but untrained model (it-bio) are indistinguishable on every general benchmark (MMLU 0.646 vs. 0.642; ARC-C 0.367 vs. 0.367; HellaSwag-norm 0.488 vs. 0.486; TruthfulQA 0.533 vs. 0.532; all ≤ 0.4 pt). Adding 28K biological tokens to the vocabulary therefore costs nothing in general competence, and any change at the BioCPT stage is attributable to the continued pretraining itself.

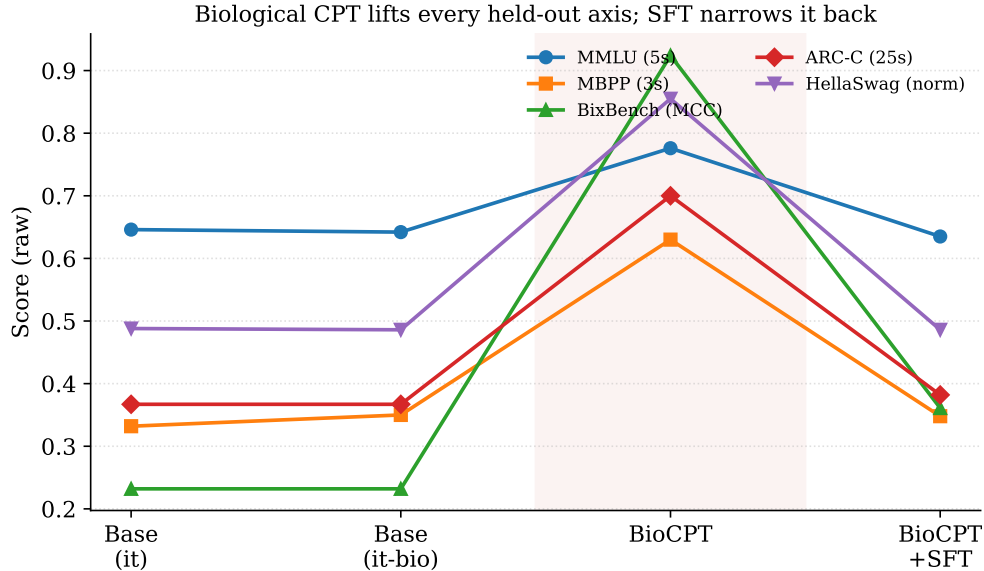


Figure 1: Every held-out capability axis follows the same trajectory: flat across the vocabulary-expansion control (it $\rightarrow$ it-bio), a sharp rise at biological CPT (shaded), and a fall back toward base after SFT. Raw scores; see Tables 1–2.

3.2 General knowledge and reasoning rise, not fall

Table 1 shows the (1) General axis. Against the intuition of catastrophic forgetting, biological CPT *improves* the model: MMLU +13.0 points (0.646 $\rightarrow$ 0.776), ARC-Challenge accuracy nearly doubles (0.367 $\rightarrow$ 0.700), and HellaSwag-norm rises +36.7 points. The one regression is TruthfulQA (−8.8 points), an interpretable drift on truthfulness/anti-misinformation after heavy exposure to sequence and literature text. We note honestly that part of the ARC/HellaSwag jump is mechanistic — an instruction-tuned model is weak at raw-loglikelihood multiple choice, and CPT on raw text restores base-model-like MC behavior — but MMLU (+13 pt, knowledge-heavy) is a hard gain unaffected by this. Subsequent SFT returns all four metrics to near-base levels.

Table 1: (1) General. Biological CPT lifts general knowledge/reasoning; SFT narrows it back. Loglikelihood multiple-choice; community-standard few-shot.

Checkpoint	MMLU (5s)	ARC-C acc (25s)	ARC-C norm	HellaSwag norm (10s)	TruthfulQA-mc2
Base (it)	0.646	0.367	0.403	0.488	0.533
Base (it-bio)	0.642	0.367	0.402	0.486	0.532
BioCPT	0.776	0.700	0.730	0.855	0.445
BioCPT+SFT	0.635	0.382	0.396	0.486	0.562

3.3 Code capability nearly doubles

Code is a structured formal language, and the (3) Coding axis (Table 2) shows the same shape. On MBPP (3-shot, the format-fair metric), pass@1 rises from 0.33 at base to **0.63** after BioCPT, then falls to 0.35 after SFT. (HumanEval 0-shot is excluded: bare completion is unfair to instruction/SFT checkpoints, which score ≈ 0 ; we report it only for completeness.) That biological-sequence CPT

196 nearly doubles Python coding ability is consistent with the hypothesis that structured-sequence
 197 pretraining strengthens a shared substrate for formal/structural languages.

Table 2: (3) Coding. MBPP 3-shot is the fair metric (HumanEval 0-shot excluded, see text).

Checkpoint	HumanEval 0s (excl.)	MBPP 3s pass@1
Base (it)	0.012	0.332
Base (it-bio)	0.000	0.350
BioCPT	0.439	0.630
BioCPT+SFT	0.000	0.348

198 3.4 Biomedical knowledge: the same curve

199 On BixBench (biomedical true/false, loglikelihood-scored MCC), the (2) Biology axis reproduces the
 200 pattern exactly: MCC 0.232 at base $\rightarrow$ **0.924** after BioCPT $\rightarrow$ 0.361 after SFT. Protein sequence-
 201 homology, by contrast, is not measurable zero-shot by any simple probe (forced Yes/No loglikelihood
 202 is near-random for all checkpoints; see §2); we read this as evidence that sequence-homology ability is
 203 *installed into representations* by CPT but only *cached out* into task performance by SFT, consistent
 204 with prior dual-mode results on this model family [8].

205 3.5 Reasoning behavior: CPT shifts the observable reasoning trajectory

206 The three axes above concern *what* the model knows. We finally ask whether CPT changes the
 207 *observable trajectory* of its chain-of-thought generations, independent of accuracy — a behavioral
 208 phenotype of the output text, not a claim about internal reasoning process. On 200 GSM8K problems,
 209 each sampled $k=5$ times under a shared 8-shot chain-of-thought scaffold, we measure generated
 210 chain length, self-correction markers (“wait/actually/however/reconsider”), hedging markers, and
 211 self-consistency (agreement of the final answer across the 5 samples). Table 3 and Figure 2 show
 212 a striking shift in this observable phenotype at the BioCPT stage: the generated chain *shortens*
 213 *by 41%* (108.8 $\rightarrow$ 64.4 tokens) and self-correction markers *nearly vanish* (0.370 $\rightarrow$ 0.006 backtracks
 214 per generation), yet accuracy does not fall (it rises slightly, 0.667 $\rightarrow$ 0.689) and self-consistency
 215 is unchanged (0.745). Biological CPT — pretraining on deterministic, low-redundancy sequence
 216 continuation — appears to shift the model’s generated text toward a *shorter, less self-revising*
 217 trajectory. Subsequent SFT reverses this: generated chains grow past base length (125.8 tokens),
 218 backtracking and hedging markers return. This is the same CPT-shapes / SFT-re-shapes signature
 219 seen on the knowledge axes, now visible in the generated reasoning *trajectory* rather than outcome
 220 — a phenotype invisible to accuracy alone. (it and it-bio again coincide, confirming the clean
 221 vocabulary ablation.)

Table 3: (4) Reasoning-behavior diagnostics on GSM8K (200 questions, $k=5$ samples, shared 8-shot CoT). CPT shortens the generated chain and removes self-revision markers without hurting accuracy; SFT reverses both.

Checkpoint	Chain len (tok)	Backtrack/gen	Hedge/gen	Self-consist.	Acc. (2nd)
Base (it)	108.8	0.370	0.001	0.749	0.667
Base (it-bio)	107.5	0.324	0.000	0.728	0.652
BioCPT	64.4	0.006	0.000	0.745	0.689
BioCPT+SFT	125.8	0.396	0.018	0.757	0.703

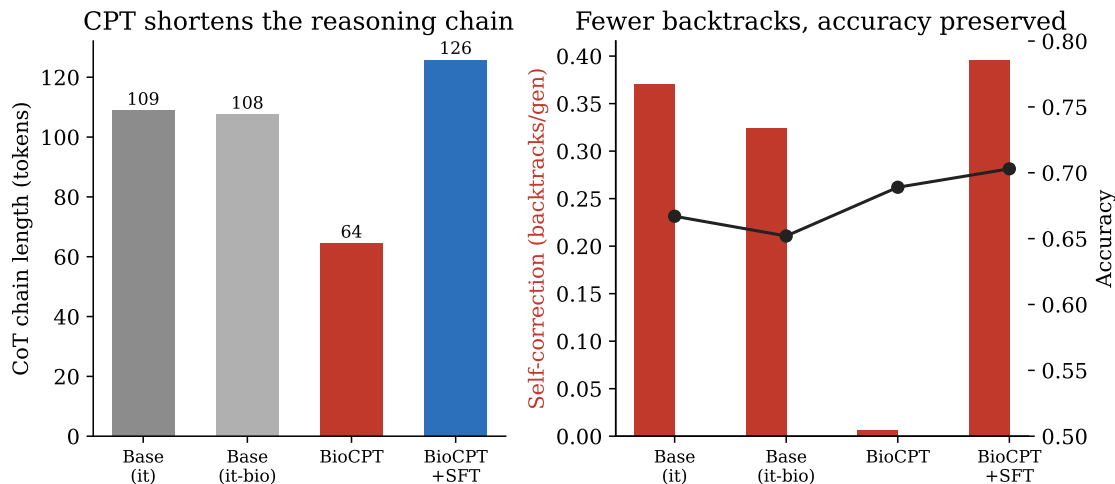


Figure 2: Observable reasoning-trajectory phenotype on GSM8K. **Left:** biological CPT shortens the generated chain-of-thought by 41%. **Right:** self-correction markers (backtracks) nearly vanish at BioCPT while accuracy is preserved (even slightly improved); SFT reverses both. CPT shifts the generated trajectory toward shorter, less revision-heavy text.

3.6 Summary

Across every completed axis the ordering is identical: base < **BioCPT** > SFT, with BioCPT the peak (Figure 3). Biological CPT is not a tax; it is an investment in a shared capability substrate. SFT then trades breadth for task focus.

4 Part II — Controlled mixture experiment

The re-analysis above is observational: it characterizes one existing lineage and cannot, on its own, establish that the *amount* and *composition* of biological data are what drive the capability shift, rather than some idiosyncrasy of that particular CPT run. We therefore ran a controlled experiment. Starting from the same vocabulary-expanded base (it-bio), we trained seven fresh CPT models under a single fixed recipe, varying only the training-data mixture, and evaluated all seven on the same axes. This turns the observational claim of §3 into a causal one: mixture ratio is a controllable lever that shapes the capability profile.

Controlled CPT protocol. Each of the seven models continues pretraining from it-bio for exactly 100M tokens (QLoRA, $r=64$ over attention, MLP, and the MoE router, with embeddings unfrozen — identical to the BioCPT recipe of §2), on a single 96 GB GPU. The *only* independent variable is the data mixture. All sources are tokenized once into per-source pools and sampled to the target ratio, so the token budget is held at 100M ($\pm 0.003\%$) across every run. Two groups vary orthogonally: (i) *Mixture ratio* — the biological share of the corpus is set to 0/5/20/50% (the remainder English web text), with the biological portion held at a fixed internal composition (70% protein, 20% DNA, 10% biomedical literature); (ii) *Composition* — the biological share is fixed at 20% and its *content* is varied (protein-only, DNA-only, protein+DNA), isolating which type of scientific data drives which capability. The 20%-mixed model (M2-bio20) serves as the shared protein+DNA+literature reference point for both groups. Evaluation reuses the general

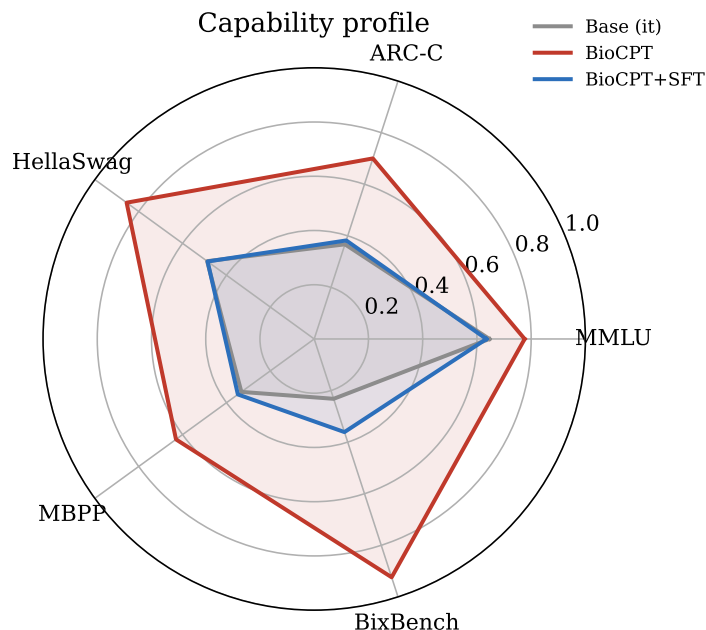


Figure 3: Capability profile of the three trained states. BioCPT (red) dominates the base model on every axis; SFT (blue) retreats toward base on all held-out axes. The biological CPT stage, not SFT, is what expands the profile.

axis (MMLU 5-shot, GSM8K 5-shot) and the biology axis (protein homology, standard and remote; BixBench), scored by loglikelihood exactly as before.

Result 1: general capability is preserved across the whole mixture range. Table 4 and Figure 4 show that raising the biological share from 0 to 50% costs essentially nothing on the general axis: MMLU moves $0.802 \rightarrow 0.796$ (-0.6 pt) and GSM8K $0.873 \rightarrow 0.861$ (-1.2 pt) — both within run-to-run noise. There is no mixture threshold at which the general axis collapses; even a corpus that is half out-of-distribution biological sequence leaves general knowledge and mathematical reasoning intact. This is the causal counterpart of the “no catastrophic forgetting” observation of §3.

Result 2: biological capability rises monotonically with biological share. Over the same sweep the biology axis climbs steadily: protein-homology discrimination (standard) rises from MCC 0.228 to 0.375 (+65%), remote homology — the harder, low-identity regime — flips from below chance to positive ($-0.023 \rightarrow +0.088$), and BixBench saturates ($0.850 \rightarrow 1.000$). The general axis is flat and the biology axis is monotone increasing over the same models: the two do not trade off. Figure 6 plots the joint trajectory, the two-dimensional form of the capability-shaping curve.

Result 3: different scientific data shape different capabilities. Holding the biological share at 20% and varying its content (Table 5, Figure 5) reveals that the *type* of scientific data matters and is not interchangeable. DNA is most effective for *remote* homology (MCC $+0.043$, versus -0.004 for protein-only), whereas protein sequence is strongest on BixBench (0.966) and standard homology;

Table 4: Controlled mixture sweep from it-bio (100M tokens each, single fixed recipe). Raising biological share preserves the general axis (MMLU, GSM8K) while monotonically lifting the biology axis. All values loglikelihood-scored; homology/BixBench are MCC.

Model	Bio%	MMLU	GSM8K	TruthQA	Homology-std	Homology-rem	BixBench
M0-base	0	0.802	0.873	0.545	0.228	−0.023	0.850
M1-bio5	5	0.801	0.867	0.524	0.270	−0.055	0.882
M2-bio20	20	0.800	0.867	0.521	0.217	−0.074	1.000
M3-bio50	50	0.796	0.861	0.519	0.375	0.088	1.000

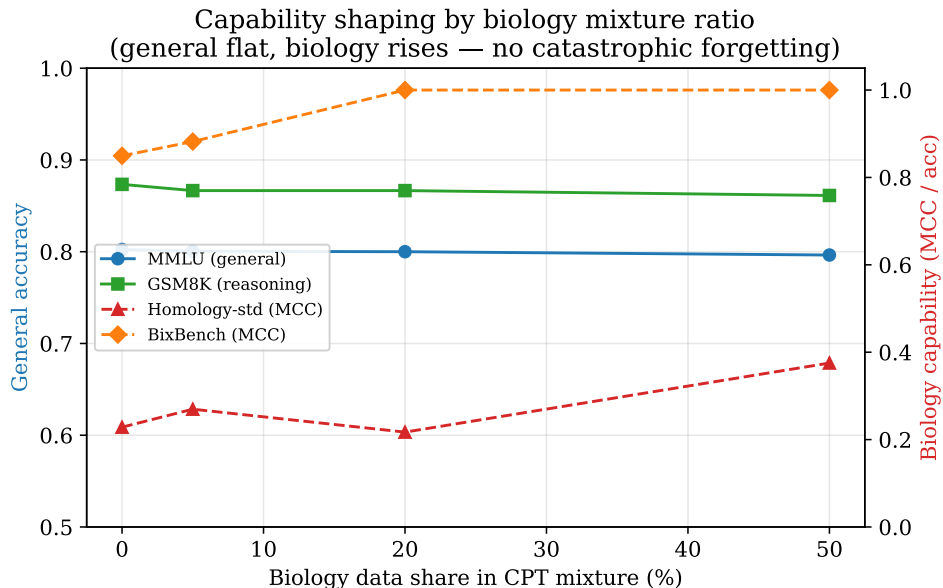


Figure 4: Capability shaping by biological mixture ratio. As the biological share of a fixed 100M-token CPT corpus rises from 0 to 50%, the general axis (MMLU, GSM8K; left scale) stays flat while the biology axis (homology MCC, BixBench MCC; right scale) rises — controlled evidence that mixture ratio shapes the capability profile without a forgetting penalty.

the protein+DNA mix is intermediate. The composition of a scientific-data corpus is thus a second, finer lever: not only *how much* scientific data, but *which* kind, selects which downstream capability is amplified.

5 Part III — A small-model, matched-compute causal probe

Part II establishes that, at 26B parameters, a biologically-mixed CPT corpus leaves the general axis essentially flat while lifting biology — the two capabilities do not trade off, but general ability is *preserved*, not *improved*. This leaves open a narrower and more provocative question, motivated directly by the forward-transfer literature this paper sits alongside [22]: biological sequence and natural-language paraphrase share a “difference-detection” structure — both require judging whether two strings that look superficially similar (or dissimilar) are functionally equivalent. If that shared structure is real, biological CPT should *measurably improve* the natural-language skills that are its direct structural counterpart, not merely leave them undisturbed. Testing this requires a regime with capability headroom, which the near-ceiling 26B model does not offer on these metrics; we

Table 5: Composition ablation at fixed 20% biological share. Different scientific-data types amplify different biological capabilities. MCC; M2-bio20 = protein+DNA+literature.

Model	Homology-std	Homology-rem	BixBench
C1-protein	0.247	−0.004	0.966
C2-dna	0.312	0.043	0.850
C3-protdna	0.237	0.025	0.914
M2-bio20 (+lit)	0.217	−0.074	1.000

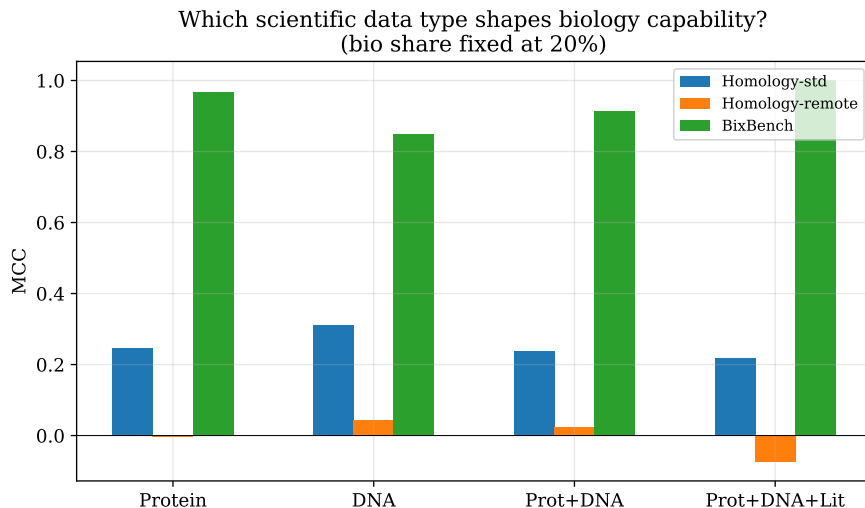


Figure 5: Composition ablation (biological share fixed at 20%). DNA most improves remote homology; protein most improves BixBench and standard homology — the type of scientific data, not just its quantity, selects which capability is shaped.

therefore turn to a small model, where headroom is available and any change is easier to attribute.

A causal, matched-compute design. We start from GPT-2 (124M parameters, 26) and continue pretraining it on English web text (openwebtext) for 300M tokens, producing a single shared checkpoint (`nl_base`). From this *one* checkpoint we then run two arms for exactly 200M further tokens each (3,051 optimizer steps, identical batch size, learning rate, and schedule) — the only difference is what the extra tokens contain: the **NL-arm** continues on openwebtext; the **Bio-arm** continues on a 50/50 protein/DNA mixture. Both arms are full-parameter finetunes (no LoRA; at 124M this is cheap and avoids any confound from adapter capacity). We repeat the two-arm comparison at three random seeds (data order and optimizer), giving a mean and standard deviation for every metric rather than a single run. Because both arms share the identical start, token budget, and optimizer trajectory, any difference between them isolates *what the extra pretraining tokens contained*, not how much training occurred.

Two families of NL probes. We evaluate both arms zero-shot on: (i) a *structural/ association* family — PAWS-en [27], the direct English counterpart of the paraphrase-detection task whose zero-shot transfer to protein homology motivated this line of work, plus the five-task character-manipulation battery from Brown et al. [28] (anagram solving on the middle letters, letter-cycling, random-letter insertion, and word-reversal), which probes sub-word structure the way homology

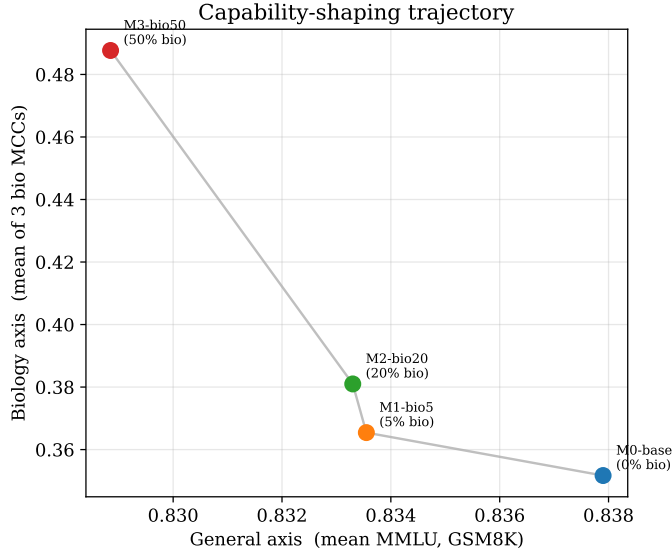


Figure 6: Two-dimensional capability-shaping trajectory of the mixture sweep. Each point is one CPT model; increasing biological share moves the model up the biology axis with negligible displacement along the general axis.

294 probes sub-sequence structure; and (ii) a *knowledge/discourse* family — HellaSwag and LAMBADA
 295 — included as a control where we expect biological data to be neutral or mildly harmful, since these
 296 tasks reward broad world-knowledge and long natural-language discourse rather than structural
 297 matching.

Table 6: Small-model matched-compute continuation (GPT-2, 200M tokens/arm, mean over 3 seeds, identical start/budget/optimizer in both arms). PAWS-en shows a reproducible biological lift; the character battery is floored for both arms at this scale (not informative); the knowledge/discourse family shows the expected cost, sharpest on long-range discourse (LAMBADA).

Family	Task	NL-arm	Bio-arm	Δ (Bio–NL), mean $\pm$ sd
Structural	PAWS-en	0.498	0.529	+0.031 $\pm$ 0.009
Structural	Anagrams (mid, 1/2)	0.000	0.000	0.000 (floor, both arms)
Structural	Cycle letters	0.000	0.000	0.000 (floor, both arms)
Structural	Random insertion	0.000	0.000	0.000 (floor, both arms)
Structural	Reversed words	0.000	0.000	0.000 (floor, both arms)
Knowledge	HellaSwag	0.291	0.285	-0.006 ± 0.000
Knowledge	LAMBADA (discourse)	0.333	0.188	-0.146 ± 0.004

298 **Result: a real, reproducible, but narrow transfer.** Table 6 shows a single clean positive
 299 result rather than a uniform one. PAWS-en accuracy rises by +3.1 points (0.498 $\rightarrow$ 0.529) in the
 300 Bio-arm, and the sign is identical across all three seeds (sd 0.009, an order of magnitude smaller
 301 than the effect) — to our knowledge the first demonstration that continuing pretraining on *raw*
 302 *biological sequence*, with no natural- language paraphrase data in the mix, measurably improves a
 303 model’s ability to detect natural- language paraphrase, under a design that holds compute exactly
 304 fixed between arms. The five character-manipulation tasks are uninformative at this scale: both
 305 arms score exactly 0, consistent with Brown et al. [28]’s own finding that these tasks are near the

noise floor below the tens-of-billions-of-parameters range; a 124M model, whether or not it has seen biological sequence, simply cannot do them, so this comparison awaits a larger base model. On the knowledge/discourse control, the cost is real: HellaSwag drops slightly (-0.6 pt) and LAMBADA drops sharply (-14.6 pt). We read the LAMBADA drop as the visible signature of a genuine trade-off at this scale that Part II’s 26B MoE model does not show: a 124M model trained fully (not via LoRA) and pushed with a 50%-biological continuation has limited spare capacity, so a half-biological corpus measurably crowds out long-range natural-language discourse modeling even as it sharpens a narrower structural skill. The dissociation is therefore partial, not clean: one structural task transfers positively and reproducibly, one discourse task pays a real cost, and the remaining structural probes are simply unmeasurable at this scale — an honest result that neither confirms nor refutes a fully general “biology helps all structure” claim, but does establish, for the first time under a causal matched-compute design, that the transfer is real for at least one task in the family motivated by this paper’s overall thesis.

6 Part IV — Robustness and mechanism

Parts I–III establish that biological CPT lifts and reshapes capability on one 26B model and transfers narrowly to natural language on one 124M model. Three questions from that evidence remain open and are addressed directly here: does the general-preserving pattern of §4 hold across model scale, not just at 26B; is uniform mixing actually forgetting anything that an explicit replay schedule could recover; and does the mixture-ratio lever operate through the same router-reorganization mechanism that full CPT uses (§3), or through something else?

6.1 Scale robustness: the general-preserving pattern holds from 2.3B to 26B

We repeat the text-only-vs-bio50 contrast of §4 (identical 100M-token budget, identical mixture, identical LoRA-CPT recipe minus the MoE router term) on two smaller, dense members of the same model family: Gemma-4-E2B (2.3B active) and Gemma-4-E4B (4.5B active), neither of which has a MoE router to unfreeze. This gives a $10\times$ range of model scale under an otherwise matched protocol.

Figure 7 shows the general-preserving property is remarkably scale-invariant: the MMLU delta between the text-only and bio-50% arms is $+0.2$, $+0.1$, and -0.6 points at E2B, E4B, and 26B respectively — all within noise, at every scale tested. The biology axis is also scale-robust on BixBench, which rises with biological share at all three sizes ($+14.4$, $+10.1$, $+15.0$ points). Homology is noisier at small scale: standard homology falls slightly with added biology at E2B and E4B (-6.5 , -2.6 points) but rises at 26B ($+14.7$ points); we read this as a capability-floor effect on weaker dense models with far fewer active parameters attending to the same pretraining signal, not as a genuine reversal, since the harder, lower-sample-count remote-homology metric is noisy in the same direction at every scale we have tested including 26B (Table 4). The honest summary is that general-preservation is established robustly across scale; biology-lift is scale-robust on the higher-sample-count BixBench task and noisier on homology at sizes well below 26B.

6.2 Replay control: uniform mixing was not forgetting anything to begin with

A natural objection to §4 is that the uniformly-shuffled bio50 mixture might be quietly trading away general capability that a smarter training schedule — for instance, an explicit block of pure text placed at the *end* of training, mimicking a replay buffer — could recover. We test this directly: holding the total token budget and the overall bio/text ratio of M3-bio50 exactly fixed, we reorder

Scale robustness of the capability-shaping pattern (2.3B $\rightarrow$ 26B, 10 $\times$ range)

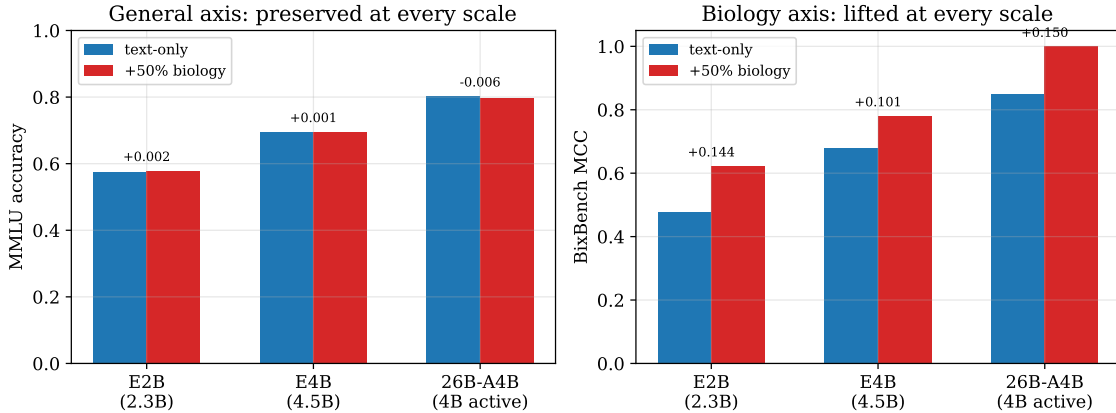


Figure 7: Text-only vs. 50%-biology contrast at three scales (E2B 2.3B, E4B 4.5B, 26B-A4B 4B active), same 100M-token budget and mixture at each size. The general axis (MMLU, left) is preserved at every scale; the biology axis (BixBench, right) is lifted at every scale.

training so the final 10% or 20% of tokens are a pure-text block (R1, R2), rather than shuffled throughout, and compare against the fully-shuffled baseline.

Table 7 and Figure 8 show the general axis is already flat across all four conditions (0.795–0.802 MMLU) — replay changes nothing on the axis it is meant to protect, because uniform shuffling was not costing anything there in the first place. On the biology axis, replay does not help and mildly *hurts*: homology-std falls monotonically as the replay fraction rises (0.375 $\rightarrow$ 0.318 $\rightarrow$ 0.266), the sensible consequence of diverting an increasing fraction of the fixed token budget away from biological content. BixBench remains near-saturated at 10% replay and dips modestly at 20%. We take this as a clean negative result rather than a null one: at this token budget and model scale, there is no forgetting problem for an explicit replay schedule to solve, and imposing one has a small opportunity cost.

Table 7: Replay control at fixed 50% biological share, 100M tokens (26B). Explicit end-of-training text replay does not change the (already-flat) general axis and mildly reduces biology-axis performance as the replay fraction rises.

Model	Replay	MMLU	GSM8K	Homology-std	BixBench
M3-bio50	0% (uniform)	0.796	0.861	0.375	1.000
R1	10% (tail)	0.800	0.868	0.318	1.000
R2	20% (tail)	0.795	0.873	0.266	0.873

6.3 Router mechanism: capability shaping and routing reorganization are dissociable at LoRA scale

§3 shows that full-parameter CPT (8.7B tokens, it-bio $\rightarrow$ BioCPT) shifts general and biology capability together; does it do so by reorganizing which experts handle which inputs? We install forward hooks on all 30 MoE routers and measure the Jensen–Shannon divergence between expert-routing distributions on protein, DNA, and natural-language prompts, for every checkpoint in this paper.

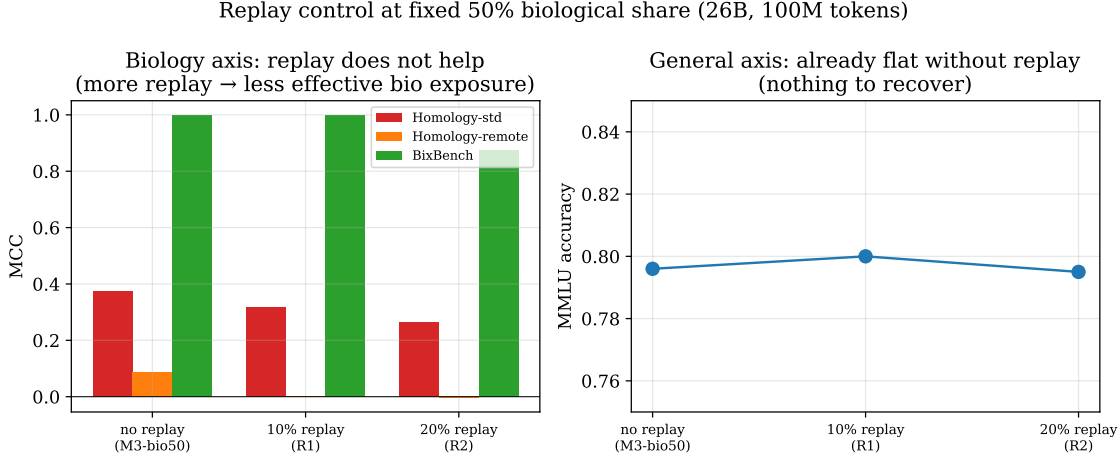


Figure 8: Replay control. Left: the biology axis does not benefit from end-of-training text replay and mildly declines as replay fraction rises. Right: the general axis is already flat without replay, so there is nothing for replay to recover.

The full-CPT lineage shows a clean, large effect: mean biology-vs-language routing divergence rises from 0.219 (it-bio, no CPT) to 0.451 (BioCPT, 8.7B tokens) — full CPT visibly reorganizes which experts serve which modality (Figure 9). The Phase-2 mixture sweep tells a different story. Despite the clear capability dose-response of §4 (homology and BixBench rising monotonically with biological share), routing divergence across M0-base through M3-bio50 is flat (0.376–0.385, no dose-response) — and M0-base (*text-only* LoRA CPT) already reaches nearly the same divergence as the 50%-biology model. The composition arm (C1–C3) shows the same flatness. In other words, at 100M-token LoRA scale, *any* continued pretraining shifts routing away from the base model by a roughly fixed amount, regardless of what the training data contains, while the *capability* gained still tracks the biological share closely. Full-scale CPT (8.7B tokens, full-parameter) is what pushes routing divergence further. We read this as evidence that capability shaping and router-level reorganization are dissociable mechanisms at the LoRA/100M-token regime, converging only at much larger training budgets — a genuine nuance rather than a clean confirmation of the naive “more biology → more specialized routing” hypothesis, and one we report as such.

7 Discussion

7.1 A division of labor between CPT and SFT

The recurring shape of our results — base < BioCPT > SFT on every held-out axis — points to a division of labor that is usually conflated. Continued pretraining does not simply add domain facts; it *re-organizes and lifts a shared capability substrate* that is then visible on tasks (general knowledge, code) far from the pretraining domain. Supervised fine-tuning does the opposite: it *narrows* the model onto the target task distribution, trading breadth on held-out axes for focus on the served task. Neither stage is a substitute for the other, and budgeting them as interchangeable — or omitting CPT to “save compute” — misreads their roles. This is consistent with, and sharpens, prior router-level evidence on this model family that CPT reshapes middle-layer computation while SFT re-aligns the output.

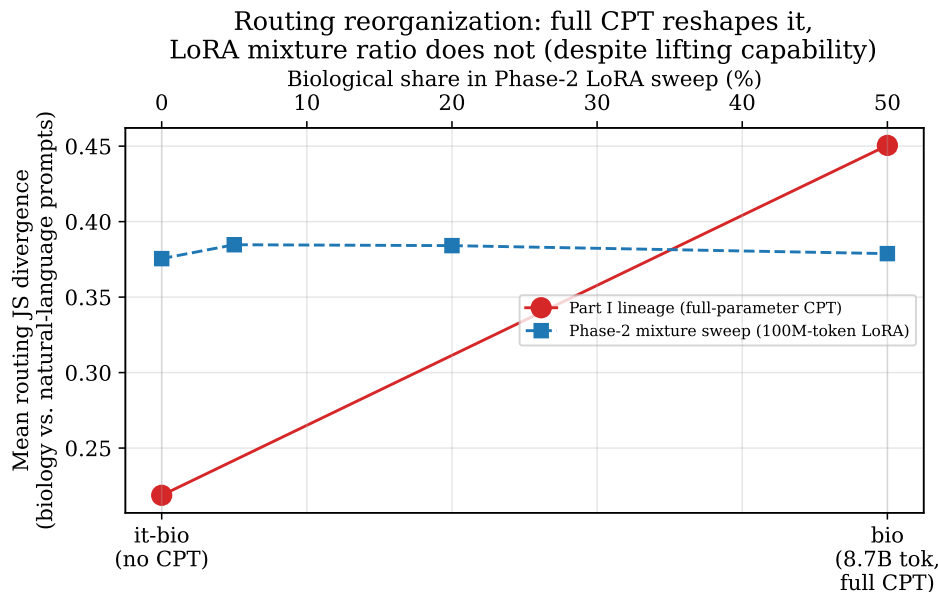


Figure 9: Routing divergence (biology-vs-language expert allocation) across the full-CPT lineage (circles, bottom axis) and the Phase-2 LoRA mixture sweep (squares, top axis). Full CPT visibly reorganizes routing; the LoRA mixture sweep does not show a dose-response in routing despite a clear one in capability.

7.2 Biological sequence as structured scientific data

If narrow, out-of-distribution biological sequence *raises* general and coding competence, then the “narrow domain = capacity competition” framing is inadequate. We propose instead that biological sequence functions as *structured scientific data*: a corpus whose value to a foundation model lies not (only) in the facts it carries but in the structural regularities it forces the model to internalize — regularities that transfer to other structured/formal languages such as code. Under this view biology is simply the first representative member of a broader class (code, mathematics, chemistry/SMILES), and the natural next question is not “does biology help biology” but “how does the *composition* of scientific-data CPT shape the capability profile of a foundation model.”

7.3 Mixture ratio is a controllable, general-preserving lever

The controlled sweep of §4 answers that question for this model. The two axes do not trade off: across a biological share of 0–50% the general axis is flat while the biology axis rises monotonically, so there is no “forgetting threshold” to budget around within this range. Equally, the composition ablation shows the lever has structure — DNA and protein amplify *different* biological capabilities — so a practitioner can, in principle, target a desired capability profile by choosing not just how much scientific data to add but which kind. This is the operational form of the data-centric thesis: training-data composition is a design surface for the capability profile, not merely a source of facts.

7.4 Capability shaping does not require routing reorganization

The router analysis of §6.3 complicates a tempting mechanistic story. It would be neat if the mixture experiment’s capability dose-response were simply the LoRA-scale echo of the routing reorganization we observe in the full-CPT lineage — more biology, more specialized experts, more biology capability. The data do not support that story: routing divergence in the mixture sweep is flat with respect to

biological share even as capability rises monotonically, and a purely *text-only* 100M-token LoRA run already produces most of the routing shift that biological runs do. What this suggests instead is that a modest LoRA-scale CPT run shifts *how* the model routes tokens in a largely data-agnostic way (any continued pretraining perturbs routing by a similar amount), while *what capability that shift buys* still depends closely on what the model was shown. Capability and routing are therefore separable readouts of continued pretraining, not two views of the same underlying change, and only converge once the training budget is large enough (here, 8.7B full-parameter tokens) to substantially reorganize the router itself. This is a useful clarification for anyone using routing metrics as a proxy for capability change under LoRA-scale adaptation: at this scale, the proxy does not track the target.

7.5 The transfer is real, but scale-dependent and narrow

Parts I and II both concern a 26B model whose general axis is near ceiling on the benchmarks we use; “preserved, not improved” is the honest description of what data composition buys there. Part III asks the sharper causal question — does biological sequence *add* natural- language capability, not merely leave it undisturbed — in a regime with headroom, and finds a real but narrow answer: paraphrase detection improves reproducibly, several other structural probes are unmeasurable at this scale, and a real cost appears in long-range discourse. This resolves an apparent tension rather than creating one. The forward-transfer hypothesis that motivates this paper — a shared difference-detection structure between natural-language paraphrase and sequence homology — predicts a *specific* transfer target, not a general uplift of all natural-language skill; PAWS-en is exactly that target, and it is the one metric that moves. That the effect appears only where capacity is scarce (a 124M full finetune) and not where it is abundant (a 26B MoE with LoRA-isolated updates) is itself informative: it suggests biological CPT’s natural-language footprint is a genuine but small-magnitude structural signal that a large, high-capacity model can absorb without displacing anything, while a small, fully-tuned model trades it against other capability. Data composition, in other words, shapes capability at every scale we tested, but what counts as a “free” addition versus a “competed-for” resource depends on how much spare capacity the training regime leaves it.

7.6 Limitations

This is a training-free re-analysis of one lineage of one 26B MoE model; the CPT / SFT recipe is fixed, so we characterize *this* lineage rather than establishing a scaling law. Part of the ARC/HellaSwag gain is a format effect (instruct-vs-completion loglikelihood), which we flag and which does not affect the knowledge-heavy MMLU gain. Zero-shot sequence-homology is not measurable by simple probes, so that axis is argued indirectly. TruthfulQA is a genuine, if small, regression and we do not hide it. The controlled experiment of §4 addresses the observational limitation directly — it fixes the recipe and varies only the data mixture — but was itself bounded: each model saw 100M CPT tokens (a pilot budget chosen to find trends, not a scaling law), on one 26B MoE model. §6.1 closes the model-size gap (the general-preserving pattern holds from 2.3B to 26B) but not the token-budget gap: whether the pattern holds at 10× the tokens, or for other scientific-data domains (code, mathematics, chemistry), remains open. Part III is deliberately narrow: one small model (GPT-2, 124M), one continuation recipe, and a character-level probe battery that is unmeasurable at this scale in either arm, so we can report only that PAWS-en transfers, not that the full structural family does; whether the effect grows, saturates, or reverses at intermediate scales between 124M and 2.3B (our smallest Part IV model) remains open. We did test whether the long-range-discourse cost of Part II’s uniform mixing could be mitigated by an explicit replay schedule (§6.2) — it could not,

because there was nothing to mitigate at this budget — but did not test whether a *lower* biological share changes that picture, nor whether Part III’s full-parameter, single-model finding generalizes across seeds and model families the way Part II’s LoRA finding does across three sizes. Finally, the router analysis of §6.3 covers only this one model family and MoE configuration; whether the LoRA/full-CPT routing dissociation is a general property of MoE continued pretraining or specific to this architecture and token regime is a question for future work.

Data and code availability

All three (four) checkpoints derive from a public lineage of Gemma-4-26B-A4B; the merged BioCPT checkpoint is released as `dnagpt/OmniGene-4-CPT-v2-merged`. Evaluation scripts (general, coding, biology-loglikelihood, and chain-of-thought diagnostics), the exact few-shot configurations, and per-example model outputs for every table in this paper are released at the project repository. All benchmarks (MMLU, ARC, HellaSwag, TruthfulQA, MBPP, HumanEval, GSM8K, BixBench) are public; no proprietary data were used. The Part I re-analysis is training-free; the seven controlled Part II CPT models (100M tokens each) together with their QLoRA adapters, data-mixture recipes, and per-example evaluation outputs are also released at the project repository. Part II reproduces on a single 96 GB GPU. The Part III GPT-2 models (one shared `n1_base` plus six matched-compute continuation checkpoints, 3 seeds $\times$ 2 arms) are full-parameter finetunes released at the same repository together with the character-manipulation probe data (mirrored from the original GPT-3-paper release) and reproduce on a single consumer-grade GPU in under two hours total. Part IV’s four scaling models (Gemma-4 E2B/E4B, text-only and 50%-biology, 100M tokens each, from the public `google/gemma-4-E2B-it` / `gemma-4-E4B-it` checkpoints) and two replay-control models (26B, 100M tokens each) are released alongside their LoRA adapters and evaluation outputs; the router-analysis code and per-checkpoint routing-divergence reports covering every checkpoint in this paper are released at the same repository. All of Part IV reproduces on two 96 GB GPUs in under two days.

Acknowledgments

Declaration of interests

The author declares no competing interests.

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